



Enhancing Transcripts of Open-Source Automatic Speech Recognition Models Through Fine-Tuning with Laughter and Speech-Laugh

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Abstract

Non-lexical sounds such as laughter are considered important discourse markers in conversation as these sounds complement the lexical information in shaping interpersonal relations, managing the conversation, and in expressing attitudes and affect. However, general purpose open-source automatic speech recognition (ASR) systems are typically not focusing on these sounds. Laughter transcription is an understudied task in ASR: laughter is often not modelled as a token and is discarded in the evaluation of ASR. In our study, we investigate how current open-source ASR models are handling laughter and speech-laugh (speech interspersed with laughter). Using Switchboard and BuckEye as conversational speech corpora, we fine-tuned and evaluated Whisper and Wav2Vec2. Our results show that laughter can be integrated in ASR transcriptions without substantially degrading word error rate.

Index Terms: automatic speech recognition, paralinguistics, laughter transcription

1. Introduction

Non-lexical sounds such as backchannels, disfluencies (e.g., fillers, hesitations) and laughter are considered important discourse markers in conversation as these complement the lexical information in turn-taking signaling, active listening, shaping and managing interpersonal relations, and in expressing attitudes and affect among others [1, 2]. Disfluencies and laughter can also be indicative of cognitive impairment and Parkinson's disease [3, 4]. However, current general purpose open-source automatic speech recognition (ASR) systems have trouble recognizing these non-lexical sounds correctly while sometimes some non-lexical sounds are discarded altogether which is generally the case with laughter [5, 6, 7, 8]. Since laughter is such an integral and important part of conversation, our aim is to investigate how the recognition of laughter can be integrated in current ASR systems.

Many researchers of human conversation such as clinicians, psychologists, legal persons, and conversational analysts rely on manual transcriptions that give detailed information about what has been said and how it has been said. ASR has the potential to support these types of transcriptions but is currently not doing this. With the exception of GailBot, which integrates existing speech-to-text services with plugins [9], current ASR systems do not produce Jefferson-style transcriptions typically used in conversation analysis (capturing nonverbal features like pause duration, laughter, and coughing) or verbatim transcriptions (preserving words exactly as spoken, without grammatical corrections and including laughter) [10]. Moreover, laughter is considered a non-lexical token (NLT) in ASR evaluations of conversational speech and is not part of the scoring (see e.g.,

NIST RT-09 Meeting Recognition Evaluation Plan [11], OpenASR21 Challenge Evaluation Plan [12]).

Laughter can be expressed in isolation, separate from speech, but is also sometimes interspersed with speech [13]. Speaking while laughing is referred to as speech-laugh. It is largely unknown to what extent words with a laughing quality can still be correctly transcribed by ASR. Hence, we will also investigate how well the speech, i.e., the words, in speech-laughs are correctly transcribed by ASR systems in conversational speech.

In this paper, the aims are to investigate 1) to what extent laughter recognition can be integrated in open-source ASR models without degrading word error rate (WER) in conversational speech, and 2) how laughing interspersed with words (speech-laughs) affects the transcription of those words. To that end, we fine-tuned and subsequently evaluated two open-source ASR models (Whisper and Wav2Vec2) using Switchboard [14] and the Buckeye Corpus [15]. We describe related work in Section 2 and the speech corpora used in Section 3. The method and results are presented in Sections 4 and 5 respectively.

2. Related work

2.1. Laughter and speech-laugh

Prior work on laughter reveals that laughter is a highly acoustically variable signal that shows quite different acoustic characteristics from speech [16, 17, 18]. Consequently, promising results have been achieved for laughter classification, 91.5% AUC [19], in the SSPNet Vocalization Corpus [20]. However, laughter detection remains quite challenging in conversational speech as [21] reports an F1 of 75.2% on Switchboard and [22] reports an F1 of 47.91% on the Buckeye Corpus. As it will be observed in Section 2.2, it remains largely unclear to what extent laughter recognition in conversational speech can be modelled jointly with ASR without degrading WER.

The extent to which the words in speech-laughs can still be transcribed by ASR systems also remains largely unknown. In Nwokah et al. [13], around 20% of all laughs in infant-directed speech were found to be co-occurring with speech. In their study, they observe that neither speech nor laughter is dominant when both are combined. F_0 of the speech-laugh seems to be determined by the frequency of speech but amplitude and rhythm may be determined by the laughter. The speech-laugh preserves some of the features of speech and laugh, but also shows the emergence of new characteristics. These observations could partly explain why speech-laughs degrade the performance of an i-vector based speaker recognition system [23], and are expected to affect transcription accuracy as well. Hence, in the current study, one of the aims is to evaluate the transcription accuracy of speech-laughs.

2.2. ASR and non-lexical tokens

The integration of recognition of NLTs in an ASR framework has gained more attention in recent years but has less often involved laughter. Most of the studies that model NLTs in ASR, address disfluent phenomena such as fillers [24, 25, 26], partial words [24], hesitations [25], reparandum and interregnum parts of disfluencies [27], or, non-disfluent short sounds in dialogue such as backchannels [24]. Horii et al. [25] tagged fillers and hesitations as # and @ and used these for training E2E models using a joint connectionist temporal classification (CTC)-attention Transformer ASR model. Error rates were reduced when disfluency labelling was applied. In Futami et al. [27], Transformer-based encoder-decoder models were trained that jointly address the speech recognition and disfluency detection task while maintaining or even improving the WER. Similarly, a study by Mihajlik et al. [24] demonstrated that WER was not negatively affected when E2E ASR acoustic models were fine-tuned on datasets with various labelling methods for disfluencies, partial words, and backchannels. Fine-tuning was also applied in a study by Zusag et al. [26] to achieve timed detection of filler events where it was also found that WER did not degrade with the addition of filler events. Overall, these studies have shown that jointly modelling NLTs such as disfluencies with (E2E) ASR models to achieve detection of these disfluencies does not degrade WER. To what extent this also holds for laughter is largely unknown.

The automatic recognition of laughter in conversation has largely been approached as an acoustic event detection [21], social signal detection [20] or non-speech information captioning task [28], rather than as part of an automated transcription task (i.e., speech-to-text conversion), although there have been some developments to unify these tasks [9, 29, 30]. Hori et al. [30] developed a meeting analyzer that monitors conversations in meetings and that includes the detection of laughter as an acoustic event that is modelled separately from the ASR module. Gail-Bot [9] combines ASR services with plugins to generate conversation analysis transcripts with paralinguistic features such as laughter. In contrast to the studies [30, 9] discussed so far, Inaguma et al. [29] present a unified framework that does integrate the detection of laughter in an ASR framework. They describe how social signal detection (which includes laughter, fillers, backchannels, and disfluencies) and ASR can be integrated in a unified framework based on connectionist temporal classification, and show that this joint modelling does not degrade ASR performance.

As previous work has demonstrated that NLTs, such as disfluencies and backchannels, can be integrated into ASR without degrading performance, we can expect the same for laughter. We will investigate how laughter transcription can be integrated in state-of-the-art open-source ASR models such as Whisper and Wav2Vec2.

3. Data

As data, we identified two conversational speech corpora that have laughter (L) and speech-laugh (SL) annotations, and that have sufficient data for our purpose.

3.1. Conversational speech corpora

For training and testing, we use the **Switchboard** Telephone Speech corpus (SWB) which contains conversational speech with over 500 speakers and over 250 hours of speech and text (transcripts) [14]. Each original audio contains two channels

referring to 2 speakers of the same conversation, separated into two audio files. The Switchboard transcripts include detailed annotations for normal text and special conventions of spontaneous conversational speech. These annotations include speakers' turns, partial words, hesitations, interruptions, vocalized sounds, such as laughter, etc. Within the laughter segments of this corpus (i.e. audio segments that contain laughter), two categories of tokens can be distinguished: laughter, marked as [laughter], and speech-laugh (i.e., words interspersed with laughter), marked as [laughter-word], where word denotes the actual word spoken.

For evaluation in a cross-corpus setting, we use the **BuckEye** Speech Corpus [31]. This corpus is another conversational speech corpus which consists of a collection of spontaneous interviews from 40 speakers from the area of Central Ohio in the US. It contains less audio than Switchboard which makes it a more challenging and low-resource dataset to test on. Within the laughter-related segments, laughter is denoted using <LAUGH>, whereas speech-laugh words are denoted as <LAUGH-word> where word denotes the actual word spoken.

3.2. Pre-processing

After a close investigation of each corpus, we noticed that several segments are noisy or silent which would negatively impact our experiments. Therefore, our pre-processing steps for the datasets include reformatting by removing silence-only and noise-only annotated sentences, as these correspond to non-speech segments, and removing or modifying special word patterns. Original audio recordings are resampled to 16kHz and clipped into multiple segments, separated by timestamps, in corresponding transcripts. The segments are shorter than 30 seconds to ensure compatibility with Whisper's architecture.

As for text, we normalized the reference transcripts (REF) and the predicted transcripts (HYP) produced by the different models. The normalization includes expanding common English contractions, removing punctuations, trailing spaces, and lowercasing the letters. We observed that disfluencies are annotated differently across the corpus, so disfluency variations were normalised to either "uh-huh" or "um-hum".

To standardize the training and evaluation of laughter, we normalize laughter tokens to <laugh> across both datasets. We consider speech-laugh as pronunciation variants of the same words spoken without laughter. Therefore, we preserve the word from [laughter-word] in Switchboard and <LAUGH-word> in BuckEye. Then we apply the same normalization techniques from the previous paragraph.

After pre-processing, Switchboard has been randomly split into 3 sets: **train**, **dev**, and **test**, with a ratio of **70%**, **10%**, and **20%** respectively. This has been achieved using Huggingface's Datasets API¹. Since the amount of data is much smaller in BuckEye, we decided to use the whole corpus as a test set only. Table 1 displays the number of tokens per dataset.

4. Methodology

4.1. Models & Training parameters

We used two pre-trained models to fine-tune the models: a version of Wav2Vec 2.0 pre-trained on 60,000 hours of speech from the Libri Vox dataset [32] and Whisper large-v2 [33]. For each pre-trained model, we have fine-tuned two versions on the **train** and **dev** set of Switchboard: one with **words**

¹<https://huggingface.co/docs/datasets/index>

Table 1: *Switchboard* and *BuckEye* token distribution. *Switchboard* is divided over train, dev and test sets. #L, #SL and #Words refer to the number of laughter tokens, speech-laughs, and words respectively.

Token	Switchboard			BuckEye
	Train	Dev	Test	Test
#L	16,044	1845	4335	1820
#SL	9586	1133	2775	1946
#Words	2,323,844	257,450	647,528	293,922
#All	2,349,474	260,428	654,638	297,688

only, laughter tokens removed ([model]-L) and one with words, speech-laugh and laughter tokens ([model]+L); here, laughter was added to the model’s tokenizer vocabulary by inserting <laugh> as a token. We also evaluated the pre-trained models’ ability to capture laughter and speech-laugh words, serving as the baselines for this research.

The training process took 20,000 steps (approximately 7 epochs) for Wav2Vec2 and 28,900 steps (approximately 10 epochs) for Whisper, with a batch size of 64, using a 0.0001 learning rate, and a weight decay of 0.005. [model]-L utilized 10% of epochs for the warmup phase while [model]+L utilized 15% of epochs with an additional linear learning rate scheduler, allowing the model to better learn the new additional token.

4.2. Experimental setup

In our experiments, we evaluated 3 different ASR models: no fine-tuning (zero-shot, Z), fine-tuned without laughter (-L), and fine-tuned with laughter (+L). As test data, we test on a distinct test set of Switchboard (within-corpus setting), and on the full BuckEye corpus (cross-corpus setting) to see how well the results transfer to a different corpus.

4.3. Metrics

To measure the ASR transcription performance, we use two versions of WER: a) WER that includes laughter in the REF and HYP transcripts, denoted by WER_{+L} , and b) WER that excludes laughter in the REF and HYP transcripts, denoted by WER_{-L} - note that this is the “standard” WER metric that discards laughter. When calculating WER_{-L} , all segments that consist of laughter only are omitted.

To measure the performance of speech-laugh transcription (i.e., transcribing the actual words in speech-laugh), we calculate the number of correctly transcribed instances as a percentage of the total number of speech-laugh (recall). To take the insertions into account as well, we used an adjusted WER, the so-called Individual Word Error Rate (IWER) [7] that can be calculated for subsets of words (instead of all words, as is the case when calculating standard WER). As a baseline, we also calculated recall and IWER for all the exact same words in speech-laugh (SL) that were *not* spoken with laughter interspersed ($\neg$ SL).

Finally, as a metric for laughter recognition performance, we use the F_1 measure with $recall = \frac{\#L_{hit}}{\#L_{REF}}$ and $precision = \frac{\#L_{hit}}{\#L_{HYP}}$.



Figure 1: WER_{+L} of Whisper and Wav2Vec2 models (zero-shot, -L: fine-tuned without laughter, +L: fine-tuned with laughter), split by the dataset that was tested on.

5. Results

5.1. Laughter

In Fig. 1, we can observe that with fine-tuning, WER_{+L} decreases overall compared to zero-shot. As expected, ASR performance is better when testing on SWB than on BuckEye, since the models were fine-tuned on SWB. Comparing Whisper to Wav2Vec2, the differences are quite small. Comparing -L to +L, we observe that adding laughter to the fine-tuning process decreases WER_{+L} performance slightly (as compared to fine-tuning without laughter) - this decrease is more apparent for Whisper than for Wav2Vec2.

More specifically, Table 2 presents both WER_{-L} and WER_{+L} and corroborates the above observations. There is improvement in transcribing the linguistic content (see WER_{-L}) when models are fine-tuned but the improvement is largest when laughter is not added as a token. The effect of adding laughter transcription on WER, however, in general is very small (see the difference between WER_{-L} and WER_{+L} for models fine-tuned with laughter). It appears that the improvement in performance of the -L models compared to the +L models can be attributed to the -L models being more accurate in transcribing the linguistic content than the +L models. Additionally, the imbalance between laughter tokens and the total word count prevents the inclusion of laughter transcription from significantly contributing to WER reduction.

Table 2: WER performances of Whisper and Wav2Vec2 models, either zero-shot (Z), fine-tuned without laughter (-L), or with laughter (+L). WER is calculated with and without laughter in the refs and hyps (WER_{+L} and WER_{-L} respectively).

	Whisper			Wav2Vec2		
	Z	-L	+L	Z	-L	+L
Test on SWB						
WER_{-L}	19.7	8.5	11.9	27.9	10.2	10.9
WER_{+L}	23.6	9.1	12.1	28.6	10.7	11.1
Test on BuckEye						
WER_{-L}	18.7	14.3	19.5	28.3	13.4	14.3
WER_{+L}	19.7	14.8	19.6	28.6	13.9	14.4

The results of laughter transcription are presented in Table 3. After fine-tuning, laughter can be transcribed in SWB and BuckEye with F_1 performances ranging from 67.5% to 79.0%, with Whisper performing better than Wav2Vec2. As

Table 3: *Laughter transcription performance in terms of Recall (R), Precision (P), F₁, deletions and substitutions (both as percentages of all laughter instances). We also show the top 5 most frequent laughter substitutions.*

	Test on SWB						Test on BuckEye					
	R	P	F ₁	%del	%sub	top 5 laughter substitutions	R	P	F ₁	%del	%sub	top 5 laughter substitutions
Whisper _Z	-	-	-	59.2	40.9	ha, bye, you, yeah, heh	-	-	-	74.0	26.0	you, ha, i, hehe, it
Whisper _L	-	-	-	58.2	41.8	uh-huh, yeah, oh, huh, uh	-	-	-	68.1	31.9	yeah, uh-huh, oh, um-hum, uh
Whisper _{+L}	80.2	77.8	79.0	11.5	8.3	yeah, know, oh, it, uh	72.3	84.4	77.9	17.7	10.0	yeah, uh, um-hum, and, know
Wav2Vec2 _Z	-	-	-	32.5	67.5	a, ha, i, e, o	-	-	-	53.9	46.0	ha, a, thats, i, ye
Wav2Vec2 _L	-	-	-	69.4	30.6	uh, h, oh, yeah, o	-	-	-	75.8	24.2	i, yeah, uh, um, um-hum
Wav2Vec2 _{+L}	78.3	77.4	77.9	14.8	6.9	i, it, t, n, o	61.8	74.5	67.5	28.4	9.8	i, you, um, know, a

expected, performances are worse for BuckEye than for SWB. When laughter is erroneously transcribed, it is more often deleted than substituted. Substitutions of laughter reveal no surprises. For Whisper, substitutions are indeed comprised of words that sound acoustically similarly to laughter, e.g., “ha, uh-huh, yeah” whereas for Wav2Vec2, substitutions are more phoneme-like, e.g., “a, ha, i, e, o”, reflecting the character-based model of Wav2Vec2.

5.2. Speech-laugh

The transcription of the words in speech-laugh is achieved with recall rates (R) (in this case, recall refers to whether the words were correctly transcribed or not) of 40.3%–70.9% and IWERs of 32.8%–70.2% (see Table 4). Whisper in general performs better than Wav2Vec2. Fine-tuning helps increase the performance of transcribing speech-laugh (for Wav2Vec2, not for Whisper) but not when laughter is added. Comparing the recall rates and IWERs between speech-laugh (SL) and the corresponding words without laughter interspersed (−SL), we can indeed observe that words interspersed with laughter are substantially worse recognized than without laughter.

Table 4: *Recall (R) and IWER (note that the lower IWER, the better) for transcription of speech-laugh (SL) and of the same speech (i.e., words) that is not interspersed with laughter (−SL). Results are split by test set (SWB, BuckEye), model (Whisper, Wav2Vec2) and model setting (zero-shot, fine-tuned without laughter: −L, fine-tuned with laughter: +L).*

Test on		zero-shot		-L		+L	
		R	IWER	R	IWER	R	IWER
SWB		Whisper					
	SL	68.0	36.0	64.5	38.6	61.6	46.7
	¬SL	81.7	19.2	94.3	7.0	91.9	9.9
BuckEye	SL	70.9	32.8	63.3	40.7	57.7	49.1
	¬SL	83.8	17.6	91.0	11.3	87.6	15.3
SWB		Wav2Vec2					
	SL	40.3	70.2	61.9	41.2	55.5	52.3
	¬SL	77.6	25.2	93.2	8.3	92.7	9.0
BuckEye	SL	56.1	46.5	57.1	46.2	55.1	49.1
	¬SL	72.4	28.7	91.1	10.9	90.6	11.6

6. Discussion

In terms of performance, our laughter recognition results are in line or even better than previous studies who have performed laughter detection on SWB and BuckEye as well: Gillick et

al. [21] and Weninger et al. [22] achieved $F_1=75.2$ and $F_1=47.9$ on SWB and BuckEye respectively where we achieved $F_1=79.0$ and $F_1=77.9$ respectively. We do find that adding laughter as a token to the fine-tuning somewhat decreases WER, which is in contrast with other studies that jointly modelled NLTs in (E2E) ASR models without degrading WER [27, 24, 26]. Similarly, we see that the performance of transcribing speech-laugh is also degraded with +L models. With precision values between 74.5% and 84.4% for laughter transcription, the introduction of laughter in the +L models cause confusion leading to higher WERs. Further inspection reveals for example that 1.6%–10.1% of all speech-laugh are erroneously confused with laughter. These observations indicate that in our study, the introduction of laughter in the fine-tuning process leads to slightly worse WERs.

For future research, we propose to add more NLTs and to investigate more models and methods for jointly modelling of these NLTs in ASR models to improve NLT transcription performance without degrading WER. In addition, fine-tuning and evaluating on more and other conversational speech corpora is needed to corroborate the current results.

7. Conclusion

We investigated to what extent laughter transcription can be integrated in Whisper and Wav2Vec2, and how the transcription accuracy of the words in speech-laugh is affected by the laughter of the speech-laugh. Through fine-tuning with laughter, we found that laughter transcription can be integrated in Whisper and Wav2Vec2 without substantially degrading WER: fine-tuning *with* laughter yielded WER_{+L} between 11.1%–19.6% while *without* laughter yielded WER_{+L} between 8.5%–14.8%. With respect to laughter recognition performance, Whisper performed slightly better than Wav2Vec2 with F_1 between 77.9%–79.0%. For speech-laugh transcription, we found that fine-tuning increases the transcription accuracy of speech-laugh for Wav2Vec2 but not for Whisper with recall rates ranging between 40.3%–68.0%. Additionally, fine-tuning *with* laughter degraded speech-laugh transcription accuracy compared to fine-tuning *without* laughter. These findings highlight the potential and limitations of integrating laughter transcription into ASR models, suggesting that while fine-tuning enables laughter transcription it may come at the cost of the accuracy of word transcription (including speech-laugh), warranting further research.

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9. References

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